Rajagiri School of Engineering & Technology (Autonomous)
Department of Electronics & Communication Engineering
B.Tech Final Year Project - Phase I | Academic Year 2026–2027
Project Phase I

Edge-Deployable AI Platform for AI-Based NeuroFusion for EEG Abnormality Measurement

Team Members	**Project Guide**
Shine Daniel — U2301192 Roshan V N — U2301179 Seban Jais — U2301190 Naveen Jose Kurian — U2301152	Ms. Harsha A

1. Introduction

Epilepsy is one of the most prevalent neurological disorders worldwide, affecting approximately 50 million people globally, with a disproportionately high burden in low- and middle-income countries. Current clinical diagnosis depends on manual interpretation of Electroencephalography (EEG) signals by trained neurologists — a process that is time-intensive, subjective, and largely inaccessible in resource-limited settings. The critical challenge in epilepsy management is not merely detection of an active preictal, but early prediction of the preictal state, which typically begins around 30 minutes before physical preictal onset. An automated and locally deployable system that can identify this preictal state would allow patients and caregivers a meaningful intervention window.

Recent advances in deep learning — particularly Temporal Convolutional Networks (TCNs) — have demonstrated that EEG brain states can be classified with acceptable accuracy. However, most published solutions depend on cloud infrastructure, introducing unacceptable latency, data privacy risks, and connectivity dependency for a continuous medical monitoring application. This project proposes and implements an edge-deployable AI platform that performs the necessary pipeline — from raw EEG signal acquisition to neural network inference — entirely on a local embedded device.

2. Objectives

- Design and implement a trained TCN network for preitcal prediction
- Develop an optimized software algorithm for feature calculation
- Deploy the trained AI model on the edge device and perform preictal prediction and classification
- interfacing of electrodes using the ADS1299EEGFE-PDK and optimize data acquisition

3. Problem Statement

Despite significant research progress, a practical gap remains between laboratory-grade EEG AI models and deployable bedside tools. The key challenges define this gap:

1. Cloud Dependency: existing AI EEG solutions require server-side computation, introducing latency and unacceptable data privacy risks for sensitive neurological data.

2. Compute Constraints: deep learning models such as TCNs are computationally demanding, requiring careful optimization before they can run on low-power edge hardware without accuracy degradation.
3. Signal Noise: raw EEG signals are heavily contaminated by electrode artifacts, muscle movement, and power line interference, requiring robust preprocessing before AI inference.

4. Proposed Solution & Methodology

The project is divided into two parallel teams with a clearly defined integration point:

Team 1 — Hardware & Signal Conditioning

- Interface the ADS1299EEG-FE daughter board directly via SPI, bypassing the MMB0 proprietary USB bridge.
- Implement digital band-pass filtering and artifact rejection on the acquired samples.
- Produce a clean, validated NxM data matrix (N channels × M samples per second) as the output deliverable.

Team 2 — AI Model & Edge Deployment

- Use the software-simulated NxM matrix (from the CHB-MIT/Bonn EEG dataset) as input, mirroring Team 1's hardware output format.
- Extract temporal and statistical features per channel over k-second sliding windows.
- Train and validate a TCN model with 6 dilated residual blocks for 4-state classification.
- Apply model pruning and quantization, then deploy via TFLite (Raspberry Pi 5).
- Benchmark real-time inference accuracy, latency, and power against published results.

5. Expected Outcomes

The following targets are set for this project:

- Implement a TCN network for preictal prediction.
- Find out the best algorithm for features extraction.
- Train the AI and get acceptable results
- Optimize and interface the electrodes with the RPi-5 with minimum errors.

6.Seminar Topics

[1] S. Al-Marzouki, "Advancing epileptic seizure recognition through bidirectional LSTM networks," Frontiers in Computational Neuroscience, vol. 19, 2025, Art. no. 1668358. doi: 10.3389/fncom.2025.1668358.

[2] P. A. Akor, G. Enemali, U. Muhammad, R. R. Singh, and H. Larijani, "An attention-residual convolutional network for real-time seizure classification on edge devices," Sensors, vol. 25, no. 22, Art. no. 6855, Nov. 2025. doi: 10.3390/s25226855.

[3] Z. Y. Lim, Y. H. Pang, S. Y. Ooi, W. H. Khoh, and Y. J. Chew, "NeuroFed-LightTCN: Federated lightweight temporal convolutional networks for privacy-preserving seizure detection in EEG data," Applied Sciences, vol. 15, no. 17, Art. no. 9660, Sep. 2025. doi: 10.3390/app15179660.

[4] L. Huang, K. Zhou, S. Chen, Y. Chen, and J. Zhang, "Automatic detection of epilepsy from EEGs using a temporal convolutional network with a self-attention layer," BioMedical Engineering OnLine, vol. 23, Art. no. 50, Jun. 2024. doi: 10.1186/s12938-024-01244-w.

7. References

[1] P. Zhu, W. Zhou, C. Cao, G. Liu, Z. Liu, and W. Shang, "A Novel SE-TCN-BIGRU Hybrid Network for Automatic Seizure Detection," *IEEE Access*, vol. 11, 2023, doi: 10.1109/ACCESS.2024.3406909.

[2] R. M. a. A. Leema, “Efficient Epileptic Seizure Detection with Optimal Channel Selection and FIXUPPACTBI-LSTM Deep Learning Model,” *IJTech - International Journal of Technology*. https://ijtech.eng.ui.ac.id/article/view/7333?hl=en-GB

[3] "Machine Learning Applied to Edge Computing and Wearable Devices for Healthcare: Systematic Mapping of the Literature," Sensors, vol. 24, no. 19, 2024. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC11478846/

[4] "Global, Regional, and National Burden of Epilepsy, 1990–2021: A Systematic Analysis for the Global Burden of Disease Study 2021," The Lancet Public Health, 2025. doi: 10.1016/S2468-2667(24)00302-5. [Online]. Available: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11876103/

[5] J. P. K. Jayalakshmi and P. Manimegalai, "Residual TCN-BiLSTM with attention framework for epileptic seizure detection from EEG signals," Int. J. Intell. Eng. Syst., vol. 19, no. 7, pp. 28–48, 2026. doi: 10.22266/ijies2026.0731.03.